BGP in IPv6

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Purpose

In this lab, I dove further into the world of BGP by exploring BGP in IPv6 and how to connect it with EIGRP and OSPF. Additionally, I learned how to use the GNS3 application system to do this lab rather than Packet Tracer to get a better sense of how to configure these things on actual routers like I used to do when I had access to actual hardware and servers.

Background Information

General Information

BGP was created in 1989 to follow its predecessor EGP (External Gateway Protocol). Unlike other routing protocols such as OSPF which splits into OSPFv2 (IPv4) and OSPFv3 (IPv6), it handles IPv4 and IPv6 together in BGP-4. One unique feature of BGP is that it does not find other BGP routers on its own and instead someone must configure the neighbor statements manually (allows BGP to run using TCP without having the concern of lost, duplicated, or rearranged messages).

BGP Improvements Over the Years

BGP version 1 was similar to EGP since it could only support hierarchical networks. A year later, this limitation was no longer a problem in BGP version 2. After another year, BGP was further refined but because it is a distance vector protocol, it continued to be less sophisticated than link-state protocols. BGP version 4 was then created in 1994 to allow for classless inter-domain routing over the Class A, B, C system and the issues that came with it. This is the version of BGP we continue to use today, but with an extension made in 1998 that is able to handle multiple protocols that was needed after the creation of IPv6. This extension is an EGP called Multiprotocol BGP and it supports most of the same features and functions of BGP in IPv4.

Setting Up IPv6 BGP

To set up IPv6 BGP, you will go into BGP like you would for IPv4, using the command "router BGP *autonomous number*" to start BGP and get into configurations for BGP. Make sure to turn off IPv4 BGP with the command "no BGP default IPv4-unicast" before setting the router ID and neighbor statements which is like the IPv4 neighbor statements that uses the IP address and autonomous number of the neighbor router. One difference between IPv4 and IPv6 however, is that your network statements are written in BGP with the command "network *IP address* mask *subnet mask*" for IPv4 while for IPv6, you don't use network statements except when configuring OSPF and EIGRP (their network statements go into specific interfaces they use). Go into your address-family with the "address-family IPv6" command and enter neighbor activate commands ("neighbor *IPv6 address* activate"), then redistribute commands for OSPF, EIGRP, and/or Connected routes. Next, you will need to set up the other routing protocols you use.

Redistribute OSPFv3 into BGP

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Activate OSPFv3 with the command “IPv6 router OSPF *autonomous number*”. Set your router ID and write the commands “redistribute connected” and “redistribute *BGP autonomous number*”. Going back to the redistribute commands in BGP’s address-family for IPv6, the commands you would use to connect routers is OSPF and BGP would be “redistribute connected” and “redistribute OSPF *autonomous number* match internal external 1 external 2”.

Redistribute IPv6 EIGRP into BGP

Start EIGRP for IPv6 with the command “IPv6 router EIGRP *autonomous number*” and proceed to set the router ID. After that, you set two redistribute commands: “redistribute connected” and “redistribute BGP *autonomous number* metric 100 1 255 1 1500”. The specific commands for EIGRP would be “redistribute EIGRP *autonomous number*” and “redistribute connected”. With this, your BGP and EIGRP routers should be connected and able to communicate.

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Lab Summary

In this lab, I worked with BGP in IPv6 on GNS3. There were seven routers and six networks total with two routers in OSPF, one in BGP, two in EIGRP and two ABRs. One ABR is between OSPF and BGP and the other is between BGP and EIGRP. I used Cisco 3660 routers for the OSPF and BGP routers (R1-4) and Cisco 7200 routers for the Area Border Router between BGP and EIGRP and the EIGRP routers (due to some issues I had with EIGRP and 3660 routers). I used Serial interfaces S1/0-S1/3 and S5/0-S5/1. My IPv6 addresses included 2001:a:: - 2001::f and 2001:aa:: networks.

Lab Commands

Address-Family IPv6: Configures the IPv6 address-family and starts the address-family configuration mode

IPv6 EIGRP #: Starts EIGRP for IPv6 on a specific interface

IPv6 OSPF # Area #: Starts OSPFv3 on a specific interface

IPv6 Router OSPF #: Starts OSPFv3 for router configuration

Neighbor *IPv6 Address* Activate: Allows neighbors to exchange IPv6 address-family prefixes

Neighbor *IPv6 Address* Remote-AS #: Adds the IPv6 address of the neighbor (identified by IPv6 address and AS number) into the IPv6 multiprotocol BGP neighbor table of the router

No BGP Default IPv4-Unicast: In the BGP protocol, this command disables the IPv4 address-family

Redistribute BGP #: Redistributes routes between routing domains (routers with different routing protocols). In this lab, this command was used for connecting to the OSPF routers from BGP routers.

Redistribute BGP # Metric 100 1 255 1 1500: Redistributes routers between routing domains (routers with different routing protocols. This specific command with the metric numbers was used to connect to EIGRP routers from BGP routers.

Redistribute Connected: Redistributes connected networks/routes (which is not commonplace but can be done indirectly and directly)

Redistribute EIGRP #: Redistributes routes from other routing protocols and domains into EIGRP

Redistribute OSPF 1 Match Internal External 1 External 2: Redistributes all OSPF routes into BGP

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Router BGP #: Starts the BGP routing process on the router

Show BGP IPv6 Neighbors: Displays IPv6 BGP neighbor connections

Show BGP IPv6 Unicast: Displays the other networks and their current status

Show BGP IPv6 Unicast Summary: Informs the administrator to see the status of the BGP connections and other related information relating to the BGP protocol

Show IPv6 Route: Shows the routing table for IPv6

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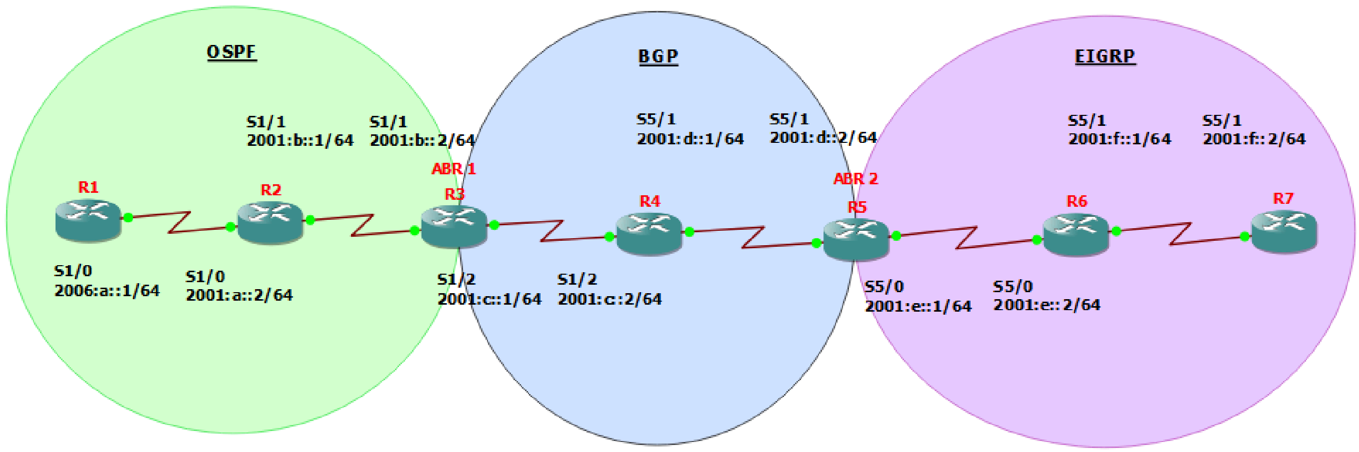
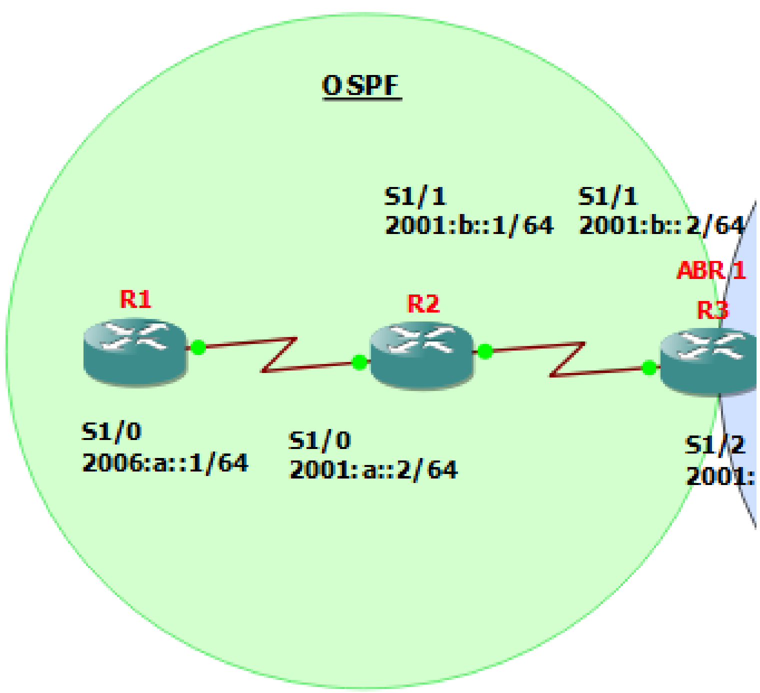
Network Diagram



OSPF



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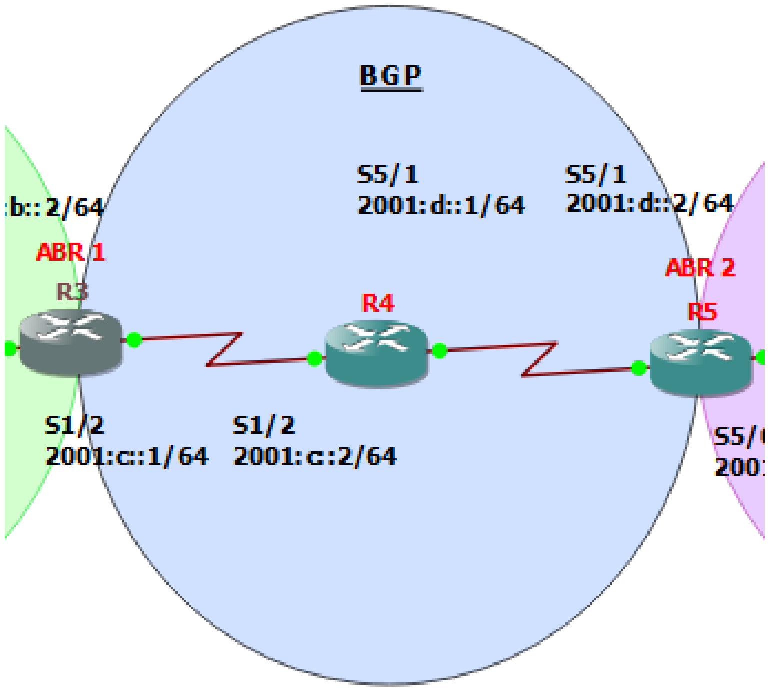
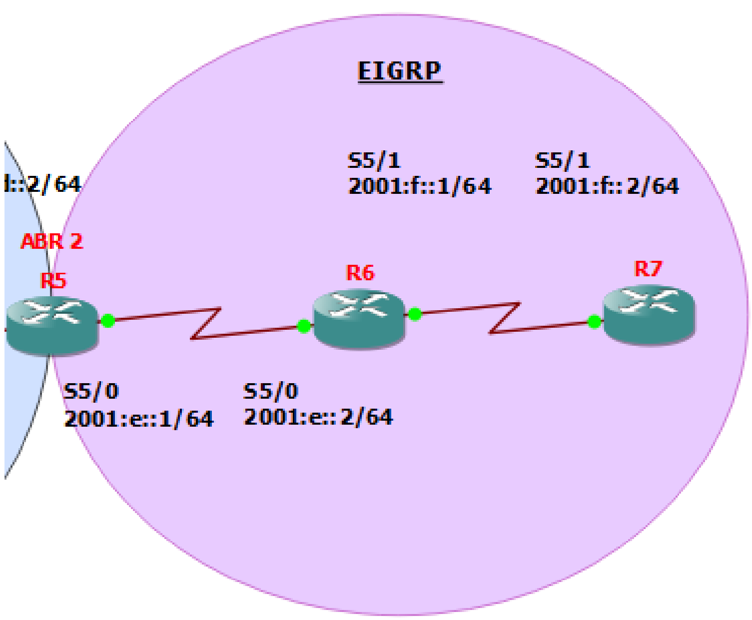
BGP



EIGRP



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Configurations

Router 1:

- **Show Run**

```
ipv6 unicast-routing

interface Serial1/0
 no ip address
 ipv6 address 2001:A::1/64
 ipv6 ospf 1 area 0
 serial restart-delay 0

ipv6 router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
```

- **Show IPv6 Route**

```
C    2001:A::/64 [0/0]
      via ::, Serial1/0
L    2001:A::1/128 [0/0]
      via ::, Serial1/0
O    2001:B::/64 [110/128]
      via FE80::CE02:CFF:FEEC:0, Serial1/0
OE2  2001:C::/64 [110/20]
      via FE80::CE02:CFF:FEEC:0, Serial1/0
OE2  2001:D::/64 [110/1]
      via FE80::CE02:CFF:FEEC:0, Serial1/0
OE2  2001:E::/64 [110/1]
      via FE80::CE02:CFF:FEEC:0, Serial1/0
OE2  2001:F::/64 [110/1]
      via FE80::CE02:CFF:FEEC:0, Serial1/0
L    FE80::/10 [0/0]
      via ::, Null0
L    FF00::/8 [0/0]
      via ::, Null0
```

- **Ping R3 (ABR 1)**
    - **S1/1 2001:b::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::2, timeout
is 2 seconds:
```

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```
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 44/72/108 ms
```

- **S1/2 2001:c::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/53/76 ms
```

- **Ping R4**
  - **S1/2 2001:c::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 36/82/108 ms
```

  - **S5/1 2001:d::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/84/116 ms
```

- **Ping R5 (ABR 2)**
  - **S5/1 2001:d::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 116/134/164 ms
```

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- **S5/0 2001:e::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:E::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 92/120/164 ms
```

- **Ping R7**
  - **S5/1 2001:f::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:F::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 152/172/200 ms
```

Router 2:

- **Show Run**

```
ipv6 unicast-routing

interface Serial1/0
 no ip address
 ipv6 address 2001:A::2/64
 ipv6 ospf 1 area 0
 serial restart-delay 0

interface Serial1/1
 no ip address
 ipv6 address 2001:B::1/64
 ipv6 ospf 1 area 0
 serial restart-delay 0

ipv6 router ospf 1
 router-id 2.2.2.2
 log-adjacency-changes
```

- **Show IPv6 Route**

```
C   2001:A::/64 [0/0]
```

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```
     via ::, Serial1/0
L    2001:A::2/128 [0/0]
     via ::, Serial1/0
C    2001:B::/64 [0/0]
     via ::, Serial1/1
L    2001:B::1/128 [0/0]
     via ::, Serial1/1
OE2  2001:C::/64 [110/20]
     via FE80::CE03:2FF:FEC0:0, Serial1/1
OE2  2001:D::/64 [110/1]
     via FE80::CE03:2FF:FEC0:0, Serial1/1
OE2  2001:E::/64 [110/1]
     via FE80::CE03:2FF:FEC0:0, Serial1/1
OE2  2001:F::/64 [110/1]
     via FE80::CE03:2FF:FEC0:0, Serial1/1
L    FE80::/10 [0/0]
     via ::, Null0
L    FF00::/8 [0/0]
     via ::, Null0
```

Router 3 (ABR 1):

- **Show Run**

```
ipv6 unicast-routing

interface Serial1/1
 no ip address
 ipv6 address 2001:B::2/64
 ipv6 ospf 1 area 0
 serial restart-delay 0

interface Serial1/2
 no ip address
 ipv6 address 2001:C::1/64
 serial restart-delay 0

router bgp 1
 bgp router-id 3.3.3.3
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 2001:C::2 remote-as 2

 address-family ipv6
  neighbor 2001:C::2 activate
  redistribute connected
```

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```
  redistribute ospf 1 match internal external 1 external 2
  no synchronization
 exit-address-family

ipv6 router ospf 1
 router-id 3.3.3.3
 log-adjacency-changes
 redistribute connected
 redistribute bgp 1
```

- **Show BGP IPv6 Neighbor**

```
BGP neighbor is 2001:C::2,  remote AS 2, external link
  BGP version 4, remote router ID 4.4.4.4
  BGP state = Established, up for 07:09:52
  Last read 00:00:52, last write 00:00:51, hold time is
180, keepalive interval is 60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(old & new)
    Address family IPv6 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
                         Sent       Rcvd
    Opens:                  2          2
    Notifications:          0          0
    Updates:                4         11
    Keepalives:          1307       1307
    Route Refresh:          0          0
    Total:               1313       1320
  Default minimum time between advertisement runs is 30
seconds

 For address family: IPv6 Unicast
  BGP table version 17, neighbor version 17/0
 Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                                 Sent       Rcvd
  Prefix activity:               ----       ----
    Prefixes Current:               3          3 (Consumes
228 bytes)
    Prefixes Total:                 3          7
    Implicit Withdraw:              0          0
    Explicit Withdraw:              0          4
    Used as bestpath:             n/a          3
```

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```
    Used as multipath:           n/a         0

                                 Outbound    Inbound
  Local Policy Denied Prefixes:  --------    -------
    AS_PATH loop:                     n/a          3
    Bestpath from this peer:            7        n/a
    Total:                              7          3
  Number of NLRIs in the update sent: max 2, min 1

  Connections established 2; dropped 1
  Last reset 07:19:37, due to Interface flap
Connection state is ESTAB, I/O status: 1, unread input
bytes: 0
Connection is ECN Disabled, Mininum incoming TTL 0,
Outgoing TTL 1
Local host: 2001:C::1, Local port: 179
Foreign host: 2001:C::2, Foreign port: 41069

Enqueued packets for retransmit: 0, input: 0  mis-ordered:
0 (0 bytes)

Event Timers (current time is 0x4FF8664):
Timer          Starts    Wakeups            Next
Retrans           451         18             0x0
TimeWait            0          0             0x0
AckHold           437        427             0x0
SendWnd             0          0             0x0
KeepAlive           0          0             0x0
GiveUp              0          0             0x0
PmtuAger            0          0             0x0
DeadWait            0          0             0x0

iss:  667261440  snduna:  667269883  sndnxt:  667269883
sndwnd:  15168
irs:  107503670  rcvnxt:  107512640  rcvwnd:      16099
delrcvwnd:    285

SRTT: 300 ms, RTTO: 303 ms, RTV: 3 ms, KRTT: 0 ms
minRTT: 16 ms, maxRTT: 328 ms, ACK hold: 200 ms
Flags: passive open, nagle, gen tcbs
IP Precedence value : 6

Datagrams (max data segment is 1440 bytes):
Rcvd: 904 (out of order: 0), with data: 439, total data
bytes: 8988
```

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```
Sent: 882 (retransmit: 18, fastretransmit: 0, partialack:
0, Second Congestion: 0), with data: 882, total data bytes:
43749
% NOTE: This command is deprecated. Please use 'show bgp
ipv6 unicast'
```

- **Show BGP IPv6 Unicast**

```
BGP table version is 17, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf
Weight Path
*> 2001:A::/64      ::                     128
32768 ?
*> 2001:B::/64      ::                       0
32768 ?
*> 2001:C::/64      ::                       0
32768 ?
*> 2001:D::/64      2001:C::2
0 2 3 ?
*> 2001:E::/64      2001:C::2
0 2 3 ?
*> 2001:F::/64      2001:C::2
0 2 3 ?
```

- **Show BGP IPv6 Unicast Summary**

```
BGP router identifier 3.3.3.3, local AS number 1
BGP table version is 17, main routing table version 17
6 network entries using 894 bytes of memory
6 path entries using 456 bytes of memory
4/3 BGP path/bestpath attribute entries using 496 bytes of
memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 1870 total bytes of memory
BGP activity 8/2 prefixes, 11/5 paths, scan interval 60
secs
```

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```
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ
Up/Down  State/PfxRcd
2001:C::2       4     2    1322    1315       17    0    0
07:11:38        3
```

- **Show IPv6 Route**

```
O   2001:A::/64 [110/128]
     via FE80::CE02:CFF:FEEC:0, Serial1/1
C   2001:B::/64 [0/0]
     via ::, Serial1/1
L   2001:B::2/128 [0/0]
     via ::, Serial1/1
C   2001:C::/64 [0/0]
     via ::, Serial1/2
L   2001:C::1/128 [0/0]
     via ::, Serial1/2
B   2001:D::/64 [20/0]
     via FE80::CE04:CFF:FE9C:0, Serial1/2
B   2001:E::/64 [20/0]
     via FE80::CE04:CFF:FE9C:0, Serial1/2
B   2001:F::/64 [20/0]
     via FE80::CE04:CFF:FE9C:0, Serial1/2
L   FE80::/10 [0/0]
     via ::, Null0
L   FF00::/8 [0/0]
     via ::, Null0
```

- **Ping R1**
  - **S1/0 2001:a::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 36/63/100 ms
```

- **Ping R4**
  - **S1/2 2001:c::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2, timeout
is 2 seconds:
```

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```
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/32/48 ms
```

- **S5/1 2001:d::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 4/28/44 ms
```

- **Ping R5 (ABR 2)**
  - **S5/1 2001:d::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/56/76 ms
```

  - **S5/0 2001:e::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:E::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/44/64 ms
```

- **Ping R7**
  - **S5/1 2001:f::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:F::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 88/122/152 ms
```

Router 4:

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- **Show Run**

```
ipv6 unicast-routing

interface Serial1/2
 no ip address
 ipv6 address 2001:C::2/64
 serial restart-delay 0

interface Serial5/1
 no ip address
 ipv6 address 2001:D::1/64
 serial restart-delay 0

router bgp 2
 bgp router-id 4.4.4.4
 no bgp default ipv4-unicast
 bgp log-neighbor-changes
 neighbor 2001:C::1 remote-as 1
 neighbor 2001:D::2 remote-as 3

 address-family ipv6
  neighbor 2001:C::1 activate
  neighbor 2001:D::2 activate
 exit-address-family
```

- **Show BGP IPv6 Neighbor**

```
BGP neighbor is 2001:C::1,  remote AS 1, external link
  BGP version 4, remote router ID 3.3.3.3
  BGP state = Established, up for 07:16:57
  Last read 00:00:56, last write 00:00:57, hold time is
180, keepalive interval is 60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(old & new)
    Address family IPv6 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                9          2
    Keepalives:           439        439
```

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```
    Route Refresh:           0           0
    Total:                 449         442
  Default minimum time between advertisement runs is 30
seconds

 For address family: IPv6 Unicast
  BGP table version 15, neighbor version 15/0
 Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                                 Sent       Rcvd
  Prefix activity:               ----       ----
    Prefixes Current:               6          3 (Consumes
228 bytes)
    Prefixes Total:                10          3
    Implicit Withdraw:              0          0
    Explicit Withdraw:              4          0
    Used as bestpath:             n/a          3
    Used as multipath:            n/a          0

                                   Outbound    Inbound
  Local Policy Denied Prefixes:    --------    -------
    Total:                                0          0
  Number of NLRIs in the update sent: max 2, min 0

  Connections established 1; dropped 0
  Last reset never
Connection state is ESTAB, I/O status: 1, unread input
bytes: 0
Connection is ECN Disabled, Mininum incoming TTL 0,
Outgoing TTL 1
Local host: 2001:C::2, Local port: 41069
Foreign host: 2001:C::1, Foreign port: 179

Enqueued packets for retransmit: 0, input: 0  mis-ordered:
0 (0 bytes)

Event Timers (current time is 0x196E62C):
Timer          Starts    Wakeups            Next
Retrans           461         15             0x0
TimeWait            0          0             0x0
AckHold           440        433             0x0
SendWnd             0          0             0x0
KeepAlive           0          0             0x0
GiveUp              0          0             0x0
PmtuAger            0          0             0x0
DeadWait            0          0             0x0
```

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```
iss:  107503670  snduna:  107512792  sndnxt:  107512792
sndwnd:  15947
irs:  667261440  rcvnxt:  667270035  rcvwnd:      15016
delrcvwnd:   1368

SRTT: 300 ms, RTTO: 303 ms, RTV: 3 ms, KRTT: 0 ms
minRTT: 4 ms, maxRTT: 324 ms, ACK hold: 200 ms
Flags: active open, nagle
IP Precedence value : 6

Datagrams (max data segment is 1440 bytes):
Rcvd: 914 (out of order: 0), with data: 440, total data
bytes: 8594
Sent: 904 (retransmit: 15, fastretransmit: 0, partialack:
0, Second Congestion: 0), with data: 904, total data bytes:
45289

BGP neighbor is 2001:D::2,  remote AS 3, external link
  BGP version 4, remote router ID 5.5.5.5
  BGP state = Established, up for 04:13:13
  Last read 00:00:22, last write 00:00:12, hold time is
180, keepalive interval is 60 seconds
  Neighbor capabilities:
    Route refresh: advertised and received(old & new)
    Address family IPv6 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0
                         Sent       Rcvd
    Opens:                  2          2
    Notifications:          0          0
    Updates:               10          8
    Keepalives:           442        483
    Route Refresh:          0          0
    Total:                454        493
  Default minimum time between advertisement runs is 30
seconds

 For address family: IPv6 Unicast
  BGP table version 15, neighbor version 15/0
 Output queue size : 0
  Index 1, Offset 0, Mask 0x2
  1 update-group member
                                 Sent       Rcvd
  Prefix activity:               ----       ----
```

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```
    Prefixes Current:               6          3 (Consumes
228 bytes)
    Prefixes Total:                 6          3
    Implicit Withdraw:              3          0
    Explicit Withdraw:              0          0
    Used as bestpath:             n/a          3
    Used as multipath:            n/a          0

                                   Outbound    Inbound
  Local Policy Denied Prefixes:    --------    -------
    Total:                                0          0
  Number of NLRIs in the update sent: max 2, min 0

  Connections established 2; dropped 1
  Last reset 04:13:25, due to Peer closed the session
Connection state is ESTAB, I/O status: 1, unread input
bytes: 0
Connection is ECN Disabled, Mininum incoming TTL 0,
Outgoing TTL 1
Local host: 2001:D::1, Local port: 179
Foreign host: 2001:D::2, Foreign port: 59149

Enqueued packets for retransmit: 0, input: 0  mis-ordered:
0 (0 bytes)

Event Timers (current time is 0x1970D88):
Timer          Starts    Wakeups            Next
Retrans           260          3             0x0
TimeWait            0          0             0x0
AckHold           280        271             0x0
SendWnd             0          0             0x0
KeepAlive           0          0             0x0
GiveUp              0          0             0x0
PmtuAger            0          0             0x0
DeadWait            0          0             0x0

iss:  176241598  snduna:  176246866  sndnxt:  176246866
sndwnd:  15453
irs:  854010769  rcvnxt:  854016365  rcvwnd:      15130
delrcvwnd:   1254

SRTT: 300 ms, RTTO: 303 ms, RTV: 3 ms, KRTT: 0 ms
minRTT: 8 ms, maxRTT: 360 ms, ACK hold: 200 ms
Flags: passive open, nagle, gen tcbs
IP Precedence value : 6

Datagrams (max data segment is 1440 bytes):
```

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```
Rcvd: 548 (out of order: 0), with data: 282, total data
bytes: 5595
Sent: 538 (retransmit: 3, fastretransmit: 0, partialack: 0,
Second Congestion: 0), with data: 538, total data bytes:
26795
% NOTE: This command is deprecated. Please use 'show bgp
ipv6 unicast'
```

- **Show BGP IPv6 Unicast**

```
BGP table version is 15, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf
Weight Path
*> 2001:A::/64      2001:C::1              128
0 1 ?
*> 2001:B::/64      2001:C::1                0
0 1 ?
*> 2001:C::/64      2001:C::1                0
0 1 ?
*> 2001:D::/64      2001:D::2                0
0 3 ?
*> 2001:E::/64      2001:D::2                0
0 3 ?
*> 2001:F::/64      2001:D::2          2681856
0 3 ?
```

- **Show BGP IPv6 Unicast Summary**

```
BGP router identifier 4.4.4.4, local AS number 2
BGP table version is 15, main routing table version 15
6 network entries using 894 bytes of memory
6 path entries using 456 bytes of memory
5/4 BGP path/bestpath attribute entries using 620 bytes of
memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2018 total bytes of memory
BGP activity 10/4 prefixes, 10/4 paths, scan interval 60
secs
```

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```
Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ
Up/Down  State/PfxRcd
2001:C::1       4     1     444     451       15    0    0
07:18:52        3
2001:D::2       4     3     495     455       15    0    0
04:14:56        3
```

- **Show IPv6 Route**

```
B   2001:A::/64 [20/128]
     via FE80::CE03:2FF:FEC0:0, Serial1/2
B   2001:B::/64 [20/0]
     via FE80::CE03:2FF:FEC0:0, Serial1/2
C   2001:C::/64 [0/0]
     via ::, Serial1/2
L   2001:C::2/128 [0/0]
     via ::, Serial1/2
C   2001:D::/64 [0/0]
     via ::, Serial5/1
L   2001:D::1/128 [0/0]
     via ::, Serial5/1
B   2001:E::/64 [20/0]
     via FE80::C805:2AFF:FE4C:0, Serial5/1
B   2001:F::/64 [20/2681856]
     via FE80::C805:2AFF:FE4C:0, Serial5/1
L   FE80::/10 [0/0]
     via ::, Null0
L   FF00::/8 [0/0]
     via ::, Null0
```

- **Ping R1**
  - **S1/0 2001:a::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 60/92/132 ms
```

- **Ping R3 (ABR 1)**
  - **S1/1 2001:b::2**

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```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 0/25/44 ms
```

- **S1/2 2001:c::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 0/23/36 ms
```

- **Ping R5 (ABR 2)**
  - **S5/1 2001:d::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/26/40 ms
```

  - **S5/0 2001:e::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:E::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 16/36/56 ms
```

- **Ping R7**
  - **S5/1 2001:f::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:F::2, timeout
is 2 seconds:
!!!!!
```

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```
Success rate is 100 percent (5/5), round-trip
min/avg/max = 60/72/92 ms
```

Router 5 (ABR 2):

- **Show Run**

```
ipv6 unicast-routing

interface Serial5/0
 no ip address
 ipv6 address 2001:E::1/64
 ipv6 eigrp 1
 serial restart-delay 0

interface Serial5/1
 no ip address
 ipv6 address 2001:D::2/64
 serial restart-delay 0

router bgp 3
 bgp router-id 5.5.5.5
 bgp log-neighbor-changes
 no bgp default ipv4-unicast
 neighbor 2001:D::1 remote-as 2

 address-family ipv6
  redistribute connected
  redistribute eigrp 1
  neighbor 2001:D::1 activate
 exit-address-family

ipv6 router eigrp 1
 eigrp router-id 5.5.5.5
 redistribute connected
 redistribute bgp 3 metric 100 1 255 1 1500
```

- **Show BGP IPv6 Neighbor**

```
BGP neighbor is 2001:D::1,  remote AS 2, external link
  BGP version 4, remote router ID 4.4.4.4
  BGP state = Established, up for 04:20:36
  Last read 00:00:35, last write 00:00:22, hold time is
180, keepalive interval is 60 seconds
  Neighbor sessions:
```

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```
    1 active, is not multisession capable (disabled)
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Four-octets ASN Capability: advertised
    Address family IPv6 Unicast: advertised and received
    Enhanced Refresh Capability: advertised
    Multisession Capability:
    Stateful switchover support enabled: NO for session 1
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

                         Sent       Rcvd
    Opens:                  1          1
    Notifications:          0          0
    Updates:                3          4
    Keepalives:           288        263
    Route Refresh:          0          0
    Total:                292        268
  Default minimum time between advertisement runs is 30
seconds

 For address family: IPv6 Unicast
  Session: 2001:D::1
  BGP table version 17, neighbor version 17/0
  Output queue size : 0
  Index 2, Advertise bit 0
  2 update-group member
  Slow-peer detection is disabled
  Slow-peer split-update-group dynamic is disabled
                                 Sent       Rcvd
  Prefix activity:               ----       ----
    Prefixes Current:               3          3 (Consumes
312 bytes)
    Prefixes Total:                 3          3
    Implicit Withdraw:              0          0
    Explicit Withdraw:              0          0
    Used as bestpath:             n/a          3
    Used as multipath:            n/a          0

                                   Outbound    Inbound
  Local Policy Denied Prefixes:    --------    -------
    AS_PATH loop:                       n/a          3
    Bestpath from this peer:              3        n/a
    Total:                                3          3
  Number of NLRIs in the update sent: max 2, min 0
  Last detected as dynamic slow peer: never
```

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```
  Dynamic slow peer recovered: never
  Refresh Epoch: 1
  Last Sent Refresh Start-of-rib: never
  Last Sent Refresh End-of-rib: never
  Last Received Refresh Start-of-rib: never
  Last Received Refresh End-of-rib: never
                                       Sent       Rcvd
        Refresh activity:              ----       ----
          Refresh Start-of-RIB          0          0
          Refresh End-of-RIB            0          0

  Address tracking is enabled, the RIB does have a route to
2001:D::1
  Connections established 2; dropped 1
  Last reset 04:21:35, due to Active open failed
  Transport(tcp) path-mtu-discovery is enabled
  Graceful-Restart is disabled
Connection state is ESTAB, I/O status: 1, unread input
bytes: 0
Connection is ECN Disabled
Mininum incoming TTL 0, Outgoing TTL 1
Local host: 2001:D::2, Local port: 59149
Foreign host: 2001:D::1, Foreign port: 179
Connection tableid (VRF): 0

Enqueued packets for retransmit: 0, input: 0  mis-ordered:
0 (0 bytes)

Event Timers (current time is 0x4E0B124):
Timer          Starts    Wakeups            Next
Retrans           295          6             0x0
TimeWait            0          0             0x0
AckHold           263        259             0x0
SendWnd             0          0             0x0
KeepAlive           0          0             0x0
GiveUp              0          0             0x0
PmtuAger        47936      47935       0x4E0B193
DeadWait            0          0             0x0
Linger              0          0             0x0

iss:  854010769  snduna:  854016517  sndnxt:  854016517
sndwnd:  14978
irs:  176241598  rcvnxt:  176246999  rcvwnd:      15320
delrcvwnd:   1064

SRTT: 300 ms, RTTO: 303 ms, RTV: 3 ms, KRTT: 0 ms
minRTT: 16 ms, maxRTT: 308 ms, ACK hold: 200 ms
```

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```
Status Flags: none
Option Flags: higher precendence, nagle, path mtu capable

Datagrams (max data segment is 1440 bytes):
Rcvd: 556 (out of order: 0), with data: 264, total data
bytes: 5400
Sent: 557 (retransmit: 6 fastretransmit: 0),with data: 557,
total data bytes: 28035


% NOTE: This command is deprecated. Please use 'show bgp
ipv6 unicast'
```

- **Show BGP IPv6 Unicast**

```
BGP table version is 17, local router ID is 5.5.5.5
Status codes: s suppressed, d damped, h history, * valid, >
best, i - internal,
              r RIB-failure, S Stale, m multipath, b
backup-path, f RT-Filter,
              x best-external, a additional-path, c
RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf
Weight Path
 *>  2001:A::/64      2001:D::1
  0 2 1 ?
 *>  2001:B::/64      2001:D::1
  0 2 1 ?
 *>  2001:C::/64      2001:D::1
  0 2 1 ?
 *>  2001:D::/64      ::                       0
32768 ?
 *>  2001:E::/64      ::                       0
32768 ?
 *>  2001:F::/64      FE80::C806:7FF:FE80:0
                                          2681856
32768 ?
```

- **Show BGP IPv6 Unicast Summary**

```
BGP router identifier 5.5.5.5, local AS number 3
BGP table version is 17, main routing table version 17
```

```
6 network entries using 1008 bytes of memory
6 path entries using 624 bytes of memory
3/3 BGP path/bestpath attribute entries using 408 bytes of
memory
1 BGP AS-PATH entries using 24 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 2064 total bytes of memory
BGP activity 10/4 prefixes, 11/5 paths, scan interval 60
secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer
InQ OutQ Up/Down  State/PfxRcd
2001:D::1       4            2     269     293       17
0    0 04:21:50        3
```

- **Show IPv6 Route**

```
B   2001:A::/64 [20/0]
     via FE80::CE04:CFF:FE9C:0, Serial5/1
B   2001:B::/64 [20/0]
     via FE80::CE04:CFF:FE9C:0, Serial5/1
B   2001:C::/64 [20/0]
     via FE80::CE04:CFF:FE9C:0, Serial5/1
C   2001:D::/64 [0/0]
     via Serial5/1, directly connected
L   2001:D::2/128 [0/0]
     via Serial5/1, receive
C   2001:E::/64 [0/0]
     via Serial5/0, directly connected
L   2001:E::1/128 [0/0]
     via Serial5/0, receive
D   2001:F::/64 [90/2681856]
     via FE80::C806:7FF:FE80:0, Serial5/0
L   FF00::/8 [0/0]
     via Null0, receive
```

- **Ping R1**
  - **S1/0 2001:a::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1, timeout
is 2 seconds:
!!!!!
```

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```
Success rate is 100 percent (5/5), round-trip
min/avg/max = 108/122/132 ms
```

- **Ping R3 (ABR 1)**
  - **S1/1 2001:b::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 44/68/80 ms
```

  - **S1/2 2001:c::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 48/60/76 ms
```

- **Ping R4**
  - **S1/2 2001:c::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 12/32/52 ms
```

  - **S5/1 2001:d::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 28/32/40 ms
```

- **Ping R7**

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- **S5/1 2001:f::2**

  ```
  Type escape sequence to abort.
  Sending 5, 100-byte ICMP Echos to 2001:F::2, timeout
  is 2 seconds:
  !!!!!
  Success rate is 100 percent (5/5), round-trip
  min/avg/max = 52/68/88 ms
  ```

Router 6:

- **Show Run**

  ```
  ipv6 unicast-routing

  interface Serial5/0
   no ip address
   ipv6 address 2001:E::2/64
   ipv6 eigrp 1
   serial restart-delay 0

  interface Serial5/1
   no ip address
   ipv6 address 2001:F::1/64
   ipv6 eigrp 1
   serial restart-delay 0

  ipv6 router eigrp 1
   eigrp router-id 6.6.6.6
  ```

- **Show IPv6 Route**

  ```
  EX  2001:A::/64 [170/26112256]
       via FE80::C805:2AFF:FE4C:0, Serial5/0
  EX  2001:B::/64 [170/26112256]
       via FE80::C805:2AFF:FE4C:0, Serial5/0
  EX  2001:C::/64 [170/26112256]
       via FE80::C805:2AFF:FE4C:0, Serial5/0
  EX  2001:D::/64 [170/2681856]
       via FE80::C805:2AFF:FE4C:0, Serial5/0
  C   2001:E::/64 [0/0]
       via Serial5/0, directly connected
  L   2001:E::2/128 [0/0]
       via Serial5/0, receive
  C   2001:F::/64 [0/0]
  ```

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```
     via Serial5/1, directly connected
L   2001:F::1/128 [0/0]
     via Serial5/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

Router 7:

- **`Show Run`**

```
ipv6 unicast-routing

interface Serial5/1
 no ip address
 ipv6 address 2001:F::2/64
 ipv6 eigrp 1
 serial restart-delay 0

ipv6 router eigrp 1
 eigrp router-id 7.7.7.7
```

- **`Show IPv6 Route`**

```
EX  2001:A::/64 [170/26624256]
     via FE80::C806:7FF:FE80:0, Serial5/1
EX  2001:B::/64 [170/26624256]
     via FE80::C806:7FF:FE80:0, Serial5/1
EX  2001:C::/64 [170/26624256]
     via FE80::C806:7FF:FE80:0, Serial5/1
EX  2001:D::/64 [170/3193856]
     via FE80::C806:7FF:FE80:0, Serial5/1
D   2001:E::/64 [90/2681856]
     via FE80::C806:7FF:FE80:0, Serial5/1
C   2001:F::/64 [0/0]
     via Serial5/1, directly connected
L   2001:F::2/128 [0/0]
     via Serial5/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
```

- **`Ping R1`**
  - **`S1/0 2001:a::1`**

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```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:A::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 184/224/368 ms
```

- **Ping R3 (ABR 1)**
  - **S1/1 2001:b::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:B::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 108/128/160 ms
```

  - **S1/2 2001:c::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 104/119/140 ms
```

- **Ping R4**
  - **S1/2 2001:c::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:C::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 64/90/104 ms
```

  - **S5/1 2001:d::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::1, timeout
is 2 seconds:
!!!!!
```

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```
Success rate is 100 percent (5/5), round-trip
min/avg/max = 76/87/100 ms
```

- **Ping R5 (ABR 2)**
  - **S5/1 2001:d::2**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:D::2, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 60/108/176 ms
```

  - **S5/0 2001:e::1**

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:E::1, timeout
is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip
min/avg/max = 56/68/100 ms
```

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Problems

For routers 5-8, I could not configure EIGRP on the Cisco 3660 routers. I was not sure what could have caused this as I have never encountered this issue before. However after some guessing and troubleshooting, I decided to use a Cisco 7200 router for any of the routes in my lab that needed EIGRP configurations. This allowed me to implement EIGRP.

After fixing the EIGRP issue, the next issue was with how to connect the 3660 and 7200 routers when the interfaces on them were different. The 3660s had interfaces S1/0-S1/3 while the 7200s serial interfaces were S5/0-S5/3. Those interfaces could not connect to each other when I tried to wire them. At first, I connected the routers through a Fast Ethernet port. I went on to finish my configurations and started pinging routers from R8 and R1 to see if both sides could ping across the networks, but R1 in OSPF could not ping part of R4 and beyond while R8 could not ping part of R5 and beyond. I looked through my configurations to see if I made a mistake and eventually came to the conclusion that using a Fast Ethernet port to connect the two different types of Cisco routers was not working. I deleted the connection and looked online for ways to connect the two. I stumbled upon an informative video on how to change the ports your router displays and are usable in GNS3. I found I had to go into the Configuration page and go to the Ports page in which I deleted all unnecessary interfaces and added interfaces S1/0-S1/3 to R5 (ABR 2). I was able to connect the two routers with the correct kind of cable now, however for some reason I could not go into any of the serial interfaces I added to the 7200 router. I decided to try it the opposite way and add interfaces S5/0-S5/1 to R4 and reconnected R4 and R5 through S5/1. I went into R4 to see if I could get into the S5 interface and I was successful. From there I reconfigured these routers and tried to ping again.

When I was pinging across the network again, I noticed I still could not ping in the same place and reviewed my configurations. Finally, I noticed my redistribute EIGRP command was in address-family IPv4. I deleted the command from address-family IPv4 as well as completely removing the address family before I added that same redistribute command to address-family IPv6 and was able to ping the part of the routers I could not before.

As I mentioned before, I could ping parts of the router I could not before, however I still could not completely ping across the network. Now R1 could not ping R6 and R8 could not ping to part of R4. Once more, I went through my configurations and talked with other peers to see what the issue was by comparing configurations. Finally, I see that my redistribute command on R5 did not have the metric at the end. My command between OSPF and BGP is “redistribute BGP 1” and I did the same command between BGP and EIGRP. I discovered that the command between those routing domains needed to be “redistribute BGP 1 metric 100 1 255 1 1500” instead. I adjusted my configurations for the last time and lo and behold, I was able to successfully ping all routers to all the other routers.

Conclusion

What I learned from this lab is how to use a new networking simulation application, GNS3, as well as more about BGP and how to configure and connect multiple routing domains in IPv6. Additionally, I learned to be more careful with my commands when doing my configurations and how to troubleshoot better in order to get my network working. I had to use multiple resources to research and troubleshoot for this lab. This can also apply to other aspects of my life and it taught me to be thorough, careful, and determined.

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